

CLUBCONNECT

PRODUCT REQUIREMENTS

DOCUMENT

Member Contribution & Recognition Platform

Hackathon Edition

Product vision • workflow • features • UI/UX • gamification • analytics • technical requirements

Based on Problem Statement 5 — ClubConnect.

1. Executive Summary

ClubConnect is a centralized platform that makes club member contributions visible, verifiable, measurable and recognized. It records technical work, event organization, mentoring, design, sponsorship, media, leadership and other club activities, then turns verified activity into points, badges, levels, analytics and a shareable contribution portfolio.

Core journey: Contribute → Submit Evidence → Verify → Earn Recognition → Build Portfolio → Generate Club Insights.

The source PS identifies the problem as contributions not always being visible or properly recorded, and calls for a transparent recognition system. It also lists gamification, backend development, data analytics, UI/UX and authentication as relevant learning areas. ■filecite■turn0file0■L59-L72■

2. Problem & Vision

Problem: Members contribute in many ways, but clubs may not have a proper record of who contributed what.

Vision: Make ClubConnect the club's single source of truth for member contribution and recognition.

Design principle: The product should immediately answer: What have I contributed? How am I progressing? What can I do next?

3. Users & Roles

Member: submits contributions, attaches evidence, tracks status, earns XP/badges/levels, joins challenges, views analytics and maintains a portfolio.

Coordinator: verifies contributions, manages members/projects/events, configures recognition rules, creates challenges and monitors engagement.

Admin: manages roles, categories, point rules, badges, club settings, access and system configuration.

4. End-to-End Workflow

New member: Landing → Sign up → Verification → Onboarding → Skills/interests → Dashboard.

Contribution: Dashboard → Add Contribution → Category → Details → Project/Event → Evidence → Preview → Submit.

Verification: Submission → Coordinator Queue → Evidence Review → Approve / Reject / Clarification → Points awarded if approved.

Recognition: Points update → Level progress → Badge rules → Challenge/streak update → Notification → Portfolio update.

Analytics: Verified contributions → Member analytics + Club analytics → trends, categories, activity and insights.

Portfolio: Verified history → projects/events/skills/badges → shareable public/private contribution profile.

5. Complete Feature Set

5.1 Landing Page

- Hero: "Every contribution deserves to be recognized."
- Get Started and Explore Club CTAs.
- Animated preview of XP, badges, contributions and leaderboard.
- How it works, contribution categories, recognition, analytics and final CTA sections.
- Responsive and accessible layout.

5.2 Authentication & Onboarding

- Login, registration, logout, password recovery and account verification.
- Optional college/organization identity or Google sign-in.
- Role-based access: Member, Coordinator, Admin.
- Onboarding wizard: basic information → skills → interests → club roles → contribution interests.

5.3 Member Dashboard

- Greeting, streak and personalized next action.
- Level + XP progress toward next level.
- Stats: XP, contributions, badges and rank.
- Recent contribution timeline with status.
- Quick actions: Add Contribution, Portfolio, Leaderboard and Challenges.

5.4 Contribution Management

- Categories: Technical, Design, Event, Marketing, Sponsorship, Mentoring, Leadership and Media.
- Title, description, date, project/event and optional effort information.
- Evidence: file upload, GitHub, website, Drive or event reference.
- Statuses: Draft, Pending, Verified, Rejected, Clarification Needed.
- Search/filter contribution history.

5.5 Contribution Verification

- Coordinator verification queue.
- Evidence preview and contributor/project context.
- Suggested/configured points shown before approval.
- Approve, reject or request clarification.
- Reviewer comments, identity, timestamp and audit history.

5.6 Points & Recognition

- Configurable point values by contribution type.
- Example: workshop +5, volunteer +10, poster +15, mentoring +20, website +30, event organization +40, major project leadership +50 XP.
- Only verified contributions award official points.
- Level progression based on cumulative XP.
- Members can see why points were awarded.

5.7 Leaderboard

- Weekly, monthly, semester and all-time views.
- Overall and category rankings.
- Technical, Design, Events, Mentoring and Leadership filters.
- Current-user rank card and distance to next rank.

- Most Improved ranking.

5.8 Badges & Achievements

- Builder, Creative, Mentor, Event Master and configurable badges.
- Milestones such as 100, 500 and 1000 XP.
- Special badges such as Consistency Champion and Multi-skilled Contributor.
- Locked badges show exact unlock requirements.
- Automatic evaluation after verified contributions.

5.9 Streaks & Challenges

- Contribution streaks.
- Weekly activity goals instead of harsh daily punishment.
- Coordinator-created challenges with goals, dates and rewards.
- Progress bars, completion states, bonus XP and challenge badges.

5.10 Analytics

- Contribution breakdown by category.
- Week/month/semester activity trends.
- Impact statistics: contributions, projects, events and mentoring.
- Contribution timeline.
- Personal progress toward levels, challenges and milestones.

5.11 Contribution Portfolio

- Shareable member contribution profile.
- Verified contributions, projects, events, skills, badges and milestones.
- Verified contribution count.
- Public/private visibility controls.
- Designed as a club achievement record.

5.12 Coordinator Dashboard

- Total members, active members, contributions and pending verification.
- Contribution trends, category distribution, member/project activity.
- Verification queue with fast actions.
- Member search and filters.
- Challenge and recognition management.

5.13 Club Health & Smart Insights

- Overall engagement status.
- Activity change over time.
- Active-member percentage.

- Top categories and areas needing attention.
- Optional AI-generated insights/recommendations.

5.14 Notifications

- Approval/rejection/clarification alerts.
- Badge unlock and level-up notifications.
- Challenge announcements and progress reminders.
- Club announcements.
- Read/unread notification center.

5.15 Search & Discovery

- Global search across members, projects and contributions.
- Filters by category, year, role, project and status.
- Coordinator member discovery.

6. Gamification System

Core loop: XP → Levels → Badges → Streaks → Challenges → Leaderboard → Recognition Portfolio.

Suggested levels: Newcomer 0–50 XP; Contributor 51–150; Active Member 151–300; Core Member 301–500; Club Leader 500+. Thresholds are configurable.

Recognition should reward meaningful work, improvement and contribution diversity—not only raw volume.

7. UI/UX Design System

Visual direction: premium, clean, modern productivity app with subtle gamification.

Style: restrained glassmorphism for hero/stat cards, transparent surfaces, backdrop blur, thin borders, soft shadows, rounded corners, subtle gradients and generous whitespace.

Readability: keep forms, tables and dense analytics surfaces more solid; avoid excessive glass.

Typography: strong metric hierarchy and highly readable body text.

Motion: short transitions, button feedback, XP gain, badge unlock and level-up animations.

Accessibility: contrast, focus states, keyboard navigation, clear labels and touch-friendly controls; never rely on color alone.

Responsive: member experience optimized for mobile; coordinator analytics optimized for desktop/tablet.

8. Screen-by-Screen UX

Dashboard: greeting → streak → level/XP → stats → recent contributions → quick actions → challenge → insight.

Add Contribution: category cards → details → evidence → preview → confirmation, with a four-step progress indicator.

Verification: queue → detail → evidence → points → reviewer action → confirmation.

Leaderboard: period tabs → category filters → podium/list → current rank → Most Improved.

Achievements: unlocked/locked gallery with progress requirements.

Portfolio: profile header → verified stats → categories → timeline → projects → badges → share/privacy.

Coordinator: KPI cards → charts → verification queue → member activity → insights → actions.

9. Information Architecture

Public: Landing, How It Works, Login, Register.

Member: Dashboard, Contributions, Add Contribution, Challenges, Achievements, Leaderboard, Analytics, Portfolio, Notifications, Settings.

Coordinator: Dashboard, Verification, Members, Contributions, Projects, Challenges, Analytics, Club Health, Notifications, Settings.

Admin: Users/Roles, Point Rules, Categories, Badges, Club Settings, Access and Audit.

10. Data Model — High Level

- User: id, name, email, role, department/year, profile, skills, interests, createdAt.
- Contribution: id, userId, categoryId, title, description, projectId/eventId, date, evidence, status, reviewerId, points, timestamps.
- Category: id, name, description, point rules.
- Project/Event: id, name, description, members/organizers, status, dates.
- Badge/UserBadge: badge criteria plus user unlock history.
- Challenge/Progress: goals, dates, reward, member progress and completion.
- Activity: userId, event type, metadata, timestamp.
- Notification: userId, type, message, read state, timestamp.
- AuditLog: actor, action, target, relevant before/after data, timestamp.

11. Business Rules & Trust

- Only verified contributions affect official points, badges and leaderboards.
- Rejected contributions award no official points.
- Clarification can be requested before final rejection.
- Point values and badge criteria are configurable by authorized users.
- Portfolio defaults to verified contributions.
- Every verification decision records reviewer and timestamp.
- Recognition-affecting actions are auditable.
- Members should understand why a contribution received its points.

12. Technical Requirements

Frontend: responsive component-based web application.

Backend: API layer for authentication, contributions, verification, points, badges, challenges, analytics and notifications.

Database: relational or document database supporting users, contributions, projects, badges and verification.

Security: secure authentication, role-based authorization and server-side recognition calculations.

Files: validated, access-controlled evidence uploads.

Analytics: optimized aggregation/query layer.

Optional AI: contribution categorization suggestions, natural-language club insights or engagement summaries. AI must not make final verification decisions.

Deployment: hosted frontend/API/database with environment-based configuration.

13. MVP vs Stretch

MVP — must work in demo: authentication/roles, member dashboard, contribution submission, evidence, coordinator verification, configurable points, levels, badges, leaderboard, basic analytics and portfolio.

Strong additions: challenges, streaks, Most Improved, timeline, Club Health and notifications.

Stretch: AI insights, AI category suggestions, QR/event check-in, project/team contribution attribution, advanced recommendations and exportable reports.

14. Hackathon Demo Workflow

- Open landing page and communicate the value proposition in 10 seconds.
- Log in as a member and show onboarding/profile.
- Show dashboard: XP, level, streak, stats and challenge.
- Submit a contribution with evidence.
- Switch to coordinator and open verification queue.
- Review evidence and approve the contribution.
- Return to member: show XP update, notification and recognition progress.
- Open leaderboard and show rank.
- Open contribution portfolio and show verified history.
- Show coordinator analytics, Club Health and smart insight.

15. Success Metrics

- Member contribution recording rate.
- Percentage of submissions verified.
- Average verification turnaround time.
- Monthly active members.
- Verified contributions per active member.
- Challenge participation.
- Recognition engagement.
- Portfolio completion rate.
- Coordinator time saved versus manual tracking.

16. UX States Checklist

- Loading/skeleton states.
- Empty states.
- Submission success confirmation.
- Pending verification.
- Rejected with reason.
- Clarification requested.
- Badge locked/unlocked.
- Challenge not started/active/completed/expired.
- Leaderboard no-data state.

- Network/error states.
- Form validation and evidence-upload errors.

17. Navigation

Mobile member: Home | Activity | + | Leaderboard | Profile.

Desktop member sidebar: Dashboard, Contributions, Challenges, Achievements, Leaderboard, Analytics, Portfolio, Notifications, Settings.

Coordinator sidebar: Dashboard, Verification, Members, Contributions, Projects, Challenges, Analytics, Club Health, Settings.

18. Product Differentiators

- Verified contribution records instead of self-reported points only.
- Contribution Portfolio that converts club work into a visible achievement record.
- Most Improved recognition for newer members.
- Club Health turns activity data into actionable insights.
- Contribution Timeline shows the complete club journey.
- Configurable recognition rather than hard-coded gamification.
- Project/team attribution keeps individual work visible.

19. Final Product Principle

ClubConnect should not feel like “a website that gives members points.” It should feel like a **verified contribution intelligence and recognition platform** where club work becomes measurable, discoverable and valuable.

Do meaningful work → Record it → Prove it → Get verified → Earn recognition → Build a portfolio → Help the club understand its people.